

A HYBRID MODELLING FRAMEWORK FOR EARLY AND LONG-TERM CEMENT STRENGTH PREDICTION

Project Id: 25-26J-250

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Department of Information Technology

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



Signature of the supervisor

29/08/2025

Date

Abstract

Cement strength is a critical parameter in the construction industry, directly influencing the durability and safety of infrastructure projects. In Sri Lanka, the demand for high-performance cement continues to grow, driven by rapid urbanization and industrial development. Despite advancements in manufacturing processes, predicting cement strength remains a complex challenge due to the variability in raw materials and production conditions. This research, sponsored by INSEE Cement, aims to address this gap by introducing a novel approach to cement strength prediction, leveraging advanced data analytics and machine learning techniques.

The proposed methodology involves the collection and analysis of comprehensive production and material datasets from INSEE's manufacturing facilities. By applying state-of-the-art machine learning models, the study seeks to identify key factors influencing cement strength and develop predictive algorithms that can accurately forecast product performance. Anticipated results include improved prediction accuracy compared to traditional methods, enabling real-time quality control and optimization of manufacturing processes. The research will also explore the integration of these predictive tools into INSEE's operational workflow, facilitating data-driven decision-making and enhancing overall product quality.

Conclusions drawn from this study are expected to demonstrate the effectiveness of machine learning in cement strength prediction, offering significant benefits to both manufacturers and end-users. Recommendations will focus on the adoption of predictive analytics for continuous process improvement and the potential for broader application across the cement industry. By pioneering this innovative approach, the project aims to set a new standard for quality assurance in Sri Lankan cement manufacturing, supporting sustainable development and infrastructure resilience.

Keywords: Cement Strength Prediction, Machine Learning, INSEE Cement, Quality Control, Sri Lanka

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1. Introduction

The construction industry is a vital driver of economic growth in Sri Lanka, with cement serving as a fundamental material for infrastructure development. As urbanization accelerates and the demand for resilient structures increases, the quality and strength of cement have become central concerns for manufacturers, builders, and policymakers alike. INSEE Cement, a leading player in Sri Lanka's cement manufacturing sector, is committed to delivering products that meet stringent performance standards and support the nation's development goals.

Despite ongoing improvements in production technology, predicting cement strength remains a significant challenge due to the complex interplay of raw material properties, manufacturing conditions, and environmental factors. Traditional quality control methods often rely on post-production testing, which can be time-consuming and may not fully capture the variability inherent in the production process. This uncertainty can lead to inefficiencies, increased costs, and potential risks in construction projects.

Recent advancements in data science and machine learning offer promising solutions to these challenges. By harnessing large-scale production data and sophisticated analytical techniques, manufacturers can gain deeper insights into the factors affecting cement strength and develop predictive models for real-time quality assurance. This research proposal outlines a novel approach to cement strength prediction, aiming to enhance process efficiency, product reliability, and industry competitiveness. The following sections will review current challenges, relevant literature, and the proposed methodology for integrating advanced analytics into Sri Lanka's cement manufacturing practices.

1.1 Background & Literature survey

Cement is a fundamental material in the construction industry, with compressive strength being a key indicator of its quality and suitability for structural applications. In Sri Lanka, the demand for high-strength cement is increasing due to rapid infrastructure development and the need for durable construction. Traditional methods for predicting cement strength rely on laboratory testing, which is time-consuming and may delay quality control decisions. Recent research has focused on leveraging machine learning and statistical techniques to predict cement strength more efficiently and accurately.

Research A:

The study "Prediction of OPC Cement Compressive Strength Based on Cement Chemical and Physical Parameters Using Machine Learning Techniques" demonstrates that models using chemical composition (e.g., CaO , SiO_2 , Al_2O_3 , Fe_2O_3) and physical parameters (Blaine fineness, particle size distribution) can predict strength with high accuracy. The integration of multiple input features and advanced algorithms enables real-time prediction, but the study is based on international datasets and does not address local production conditions [1].

Research B:

"Strength Prediction and Optimisation of Velvet Tamarind Pod Ash Cement Blends via Response Surface Methodology" explores the use of alternative materials and statistical optimization. Response surface methodology effectively models and optimizes the compressive strength of blended cements, but is limited in handling complex, nonlinear relationships compared to machine learning [2].

Research C:

"Analysis and prediction of compressive strength of calcium aluminate cement paste based on machine learning" applies machine learning to specialty cements,

demonstrating improved prediction accuracy and the ability to identify key influencing factors. However, the study focuses on calcium aluminate cement and does not address the broader context of ordinary Portland cement (OPC) production in Sri Lanka [3].

Key Findings:

- Machine learning models outperform traditional statistical methods in strength prediction.
- Comprehensive input data (chemical, physical, and process parameters) are essential for accurate predictions.
- Most studies are based on international datasets and do not address local production conditions or real-time application in Sri Lankan cement plants.

Proposed Novelty Implementation: Real-Time Cement Strength Prediction and Process Optimization Platform

Function Overview:

This novel system integrates real-time plant and laboratory data—including chemical composition, Blaine fineness, gypsum content, and process conditions—into a machine learning-powered prediction engine. The platform provides instant strength estimates at 2, 7, and 28 days, highlights key influencing factors, and issues early warnings if predicted strength falls below target. Operators can adjust raw mix, grinding fineness, or gypsum content proactively, optimizing quality before full lab results are available.

Key Features:

1. Real-Time Strength Prediction:
 - Analyses comprehensive input data (chemical, physical, process, and environmental) using advanced machine learning models.
 - Predicts compressive strength at multiple curing ages for each cement batch.

2. Actionable Feedback:

- Identifies key factors affecting strength (e.g., low Blaine, optimum SO₃).
- Provides early warnings and recommendations for process adjustments.

3. Local Validation:

- Tailored to Sri Lankan OPC production, validated with INSEE Cement's operational data.

4. Process Optimization:

- Enables plant operators and QC engineers to optimize production parameters in real time, improving efficiency and product reliability.

This innovative approach empowers Sri Lankan cement manufacturers to move from reactive, post-production testing to proactive, data-driven quality control and process optimization, setting a new standard for the industry.

1.2 Research Gap

Despite significant advancements in cement strength prediction using machine learning and statistical methods, a substantial gap remains in the practical application of these technologies within the Sri Lankan cement industry. Most existing studies are based on international datasets, focus on specialty cements, or utilize post-production laboratory results, limiting their relevance and impact for local manufacturers such as INSEE Cement.

Current research does not address the integration of real-time plant and laboratory data including chemical composition, Blaine fineness, gypsum content, and process condition into predictive models tailored for ordinary Portland cement (OPC) produced in Sri Lanka. Furthermore, environmental factors such as curing temperature and humidity, which can significantly influence strength outcomes, are often overlooked or inadequately incorporated.

Another critical gap is the lack of actionable feedback and early warning systems. Existing models typically provide strength predictions without identifying key influencing factors or offering recommendations for process adjustments. This limits the ability of plant operators and quality control engineers to proactively optimize production parameters and ensure consistent product quality.

Additionally, there is an absence of platforms validated with local operational data that can deliver rapid, batch-specific strength estimates at multiple curing ages (2, 7, and 28 days). Without such tools, Sri Lankan manufacturers must rely on time-consuming laboratory tests, resulting in delays, inefficiencies, and increased risk of non-compliance with quality standards.

Addressing these gaps requires the development of a novel, real-time prediction system that leverages advanced machine learning techniques, integrates comprehensive production and environmental data, and provides actionable insights for process optimization. Such a solution would empower Sri Lankan cement manufacturers to transition from reactive quality control to proactive, data-driven decision-making, setting a new benchmark for efficiency and reliability in the industry.

Feature/Function	Research A	Research B	Research C	Proposed System
Real-Time Strength Prediction	✓	✗	✓	✓
Integration of Chemical & Physical Data	✓	✓	✓	✓
Process/Environmental Corrections	✗	✓	✗	✓
Early Warning for QC	✗	✗	✗	✓
Application to Sri Lankan OPC	✗	✗	✗	✓

Table 1: Comparison of Existing Research and Proposed System

Upon reviewing the existing research [1] [2] [3], it is clear that significant gaps remain in addressing the practical needs of Sri Lankan cement manufacturers, especially regarding real-time strength prediction and process optimization. While previous studies have demonstrated the effectiveness of machine learning and statistical methods for predicting compressive strength, none offer integrated solutions that combine real-time plant and laboratory data, environmental corrections, and actionable feedback for quality control.

The comparative table highlights that existing research typically focuses on either chemical and physical data integration or statistical optimization but lacks features such as early warning systems and direct application to Sri Lankan OPC production. In contrast, the proposed system incorporates all these elements, providing real-time strength estimates, key influencing factor analysis, and immediate recommendations for process adjustments. This comprehensive approach enables proactive quality control and process optimization, setting a new benchmark for the local cement industry.

By bridging these gaps, the proposed platform offers a novel and highly beneficial solution, empowering manufacturers like INSEE Cement to enhance efficiency, reliability, and product quality beyond what current research provides.

1.3 Research Problem

The Sri Lankan cement industry is a cornerstone of national infrastructure development, yet the accurate and timely prediction of cement compressive strength remains a significant challenge. Traditional methods rely on post-production laboratory testing, which can be slow and resource-intensive, often resulting in delays in quality assurance and process optimization. This approach limits the ability of manufacturers to make real-time adjustments to production parameters, increasing the risk of producing batches that do not meet required standards.

The core issue is how to enhance the accuracy and efficiency of cement strength prediction using modern technological solutions. This includes addressing challenges related to integrating real-time chemical, physical, and process data, applying environmental corrections, and providing actionable feedback for quality control. Furthermore, the lack of early warning systems and batch-specific recommendations hinders proactive process optimization and increases operational costs.

To address these challenges, the research will focus on developing advanced machine learning models that utilize comprehensive production and laboratory data to predict compressive strength at multiple curing ages. The proposed system will deliver instant strength estimates, highlight key influencing factors, and provide early warnings and recommendations for process adjustments. By enabling real-time, data-driven decision-making, this research aims to transform quality control and process optimization in Sri Lankan cement manufacturing, ensuring consistent product reliability and operational efficiency.

2. Objectives

2.1. Main Objective

To develop a real-time, data-driven prediction system that leverages advanced machine learning techniques to accurately forecast the compressive strength of ordinary Portland cement (OPC) produced in Sri Lanka, enabling proactive quality control and process optimization for INSEE Cement.

2.2 Specific Objectives

2.2.1 Real-Time Strength Prediction

Objective:

To provide instant, batch-specific predictions of cement compressive strength at 2, 7, and 28 days using comprehensive chemical, physical, and process data.

Details:

- Integrate real-time plant and laboratory data (oxide analysis, Blaine fineness, gypsum content, process conditions).
- Apply machine learning models to predict strength at multiple curing ages.
- Validate predictions against laboratory results for accuracy.

2.2.2 Actionable Quality Control Feedback

Objective:

To deliver early warnings and actionable recommendations for process adjustments when predicted strength falls below target thresholds.

Details:

- Identify key influencing factors (e.g., low Blaine, suboptimal SO_3).
- Provide feedback to plant operators and QC engineers for immediate corrective actions.
- Reduce reliance on post-production laboratory testing.

2.2.3 Process Optimization and Efficiency

Objective:

To enable plant operators to optimize raw mix, grinding fineness, and gypsum content in real time, improving production efficiency and product reliability.

Details:

- Use model outputs to guide adjustments in production parameters.
- Minimize waste and rework by ensuring batches meet quality standards before lab results are available.
- Support continuous improvement through data-driven insights.

2.2.4 Support for R&D and New Product Development

Objective:

To facilitate rapid evaluation of alternative raw materials and supplementary cementitious materials (SCMs) for new product development.

Details:

- Simulate the impact of different raw material blends on compressive strength.
- Accelerate R&D cycles by reducing the need for extensive laboratory experiments.
- Support innovation in sustainable cement formulations.

2.2.5 Local Validation and Industry Adoption

Objective:

To ensure the system is tailored to Sri Lankan production conditions and validated with INSEE Cement's operational data for industry-wide adoption.

Details:

- Collect and analyze local production and environmental data.
- Collaborate with INSEE Cement for pilot implementation and feedback.

3. Methodology

The methodology for developing the real-time cement strength prediction and process optimization system for INSEE Cement consists of several structured stages. This approach combines user-centered design, advanced machine learning, and integration with existing plant operations to ensure the system is practical, accurate, and impactful for Sri Lankan cement manufacturing.

3.1 System Architecture Diagram

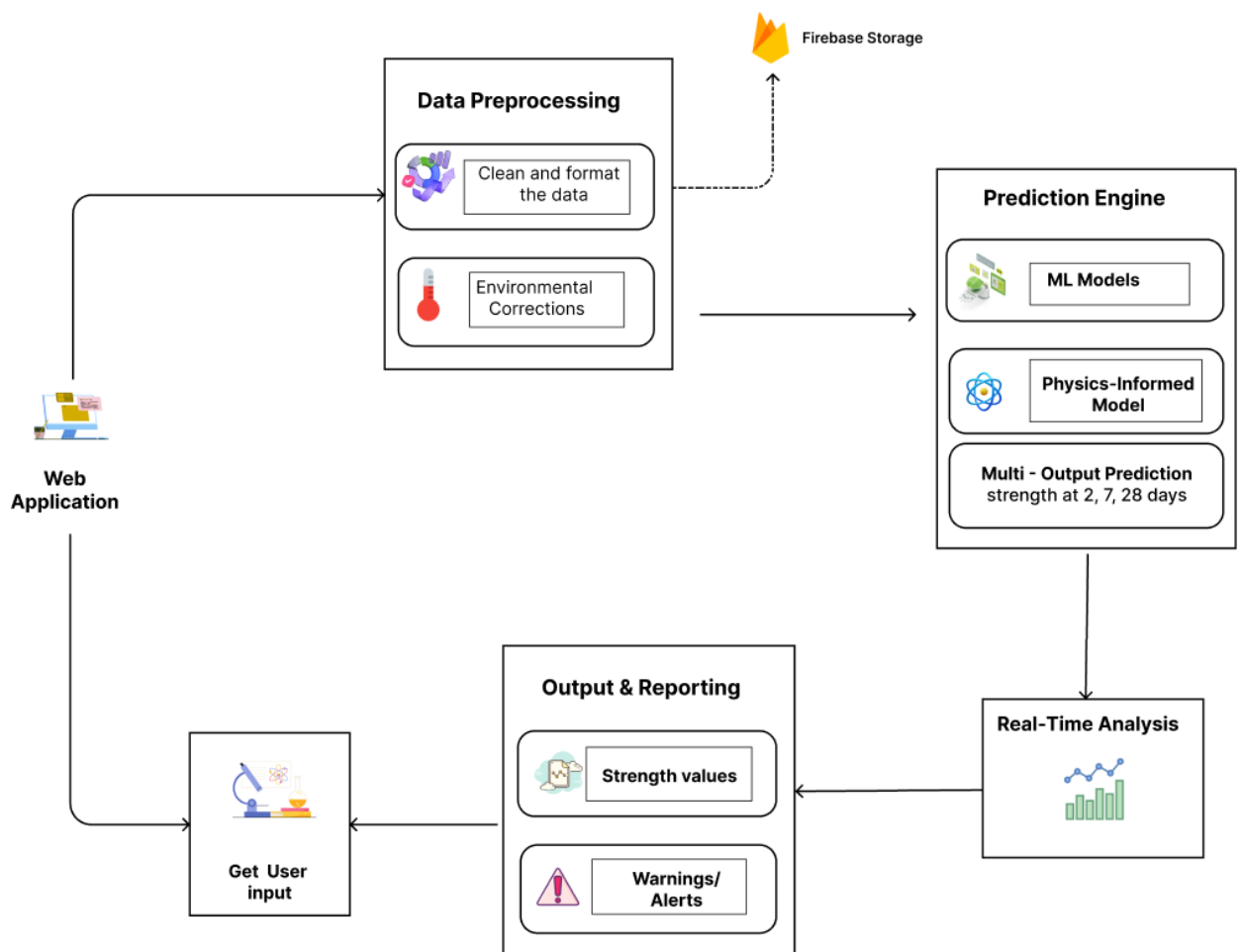


Figure 1: Function Overview

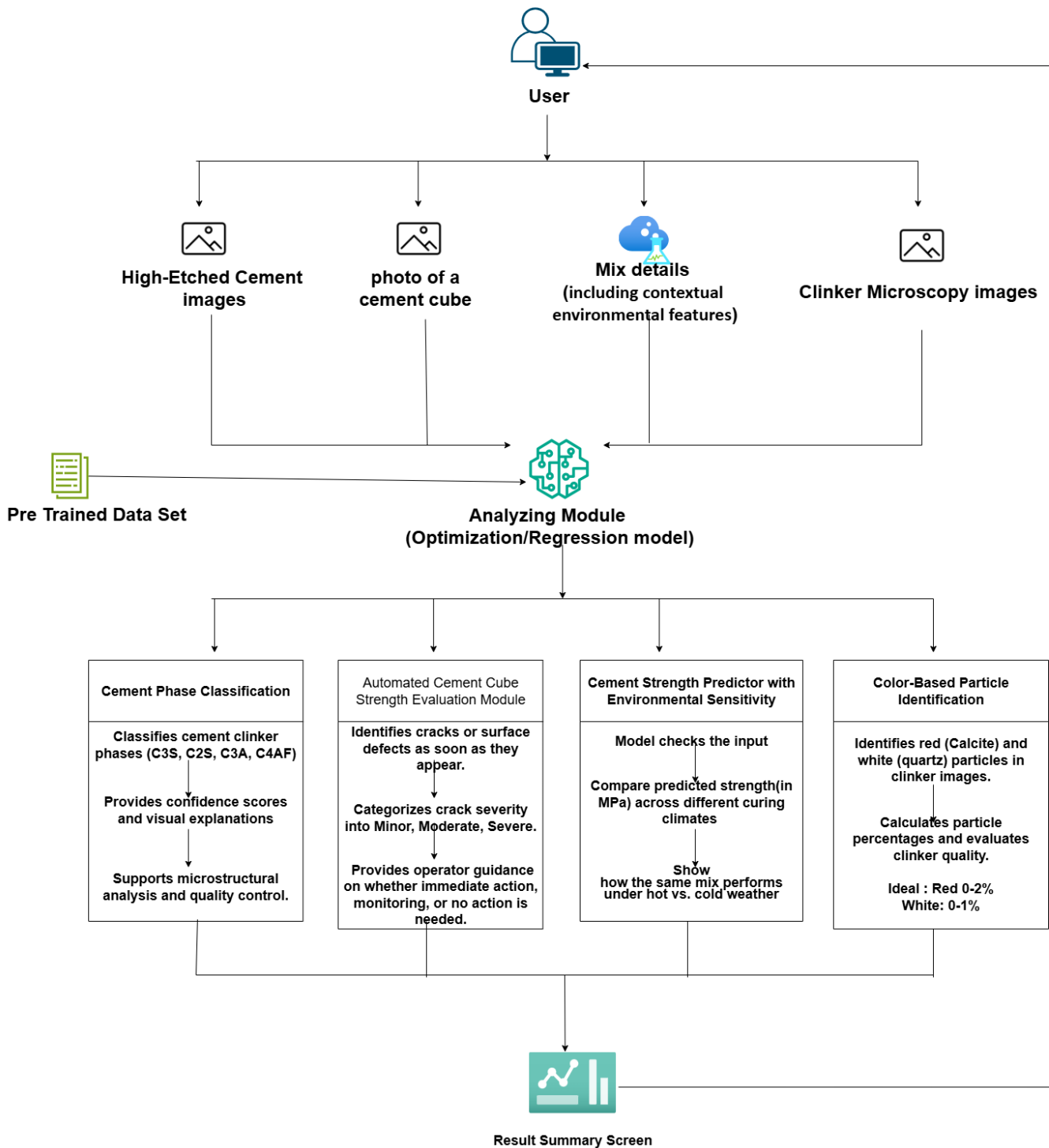


Figure 2: System Overview

3.2 Understanding the Main Components of the Research Area

3.2.1 Data Input System

The prediction process begins with the collection of cement batch data from the quality control (QC) laboratory and plant operations. A web application will be developed as the interface for data entry.

The system allows QC staff to enter critical input variables such as cement chemistry (oxide composition, Bogue phases), fineness (Blaine and PSD values), gypsum content, supplementary cementitious material (SCM) percentage, and environmental factors (curing temperature, humidity, CO₂ exposure).

This structured input ensures that all essential features influencing cement strength are captured consistently. Once entered, the data is transmitted to the system server for initial preprocessing.

3.2.2 Data Preprocessing

The preprocessing stage ensures the input dataset is clean, accurate, and suitable for predictive analysis.

- **Noise Reduction & Cleaning:** Incorrect or missing entries (e.g., missing oxide values) are flagged and corrected.
- **Normalization:** Features such as Blaine fineness, oxide percentages, and curing age are standardized to maintain consistency.
- **Feature Engineering:** New variables (e.g., $C_3S \times \text{Blaine}$ interaction, SO_3^{2-} , hydration rate factors) are generated to capture hidden relationships.
- **Environmental Adjustment:** Correction factors are applied to account for curing temperature, humidity, and CO₂ conditions that impact hydration and strength development.

3.2.3 Strength Prediction Engine

This is the core analytical component of the system. The processed dataset is passed into hybrid models designed to predict cement strength at 2, 7, and 28 days.

- **Physics-Informed Model:** A gel-space ratio and hydration kinetics model is implemented for explainability and alignment with cement hydration theory.
- **Machine Learning Model:** Gradient Boosting, Random Forest, and Neural Networks are trained on historical plant datasets for accuracy.
- **Multi-Output Prediction:** The models simultaneously predict compressive strength values at multiple ages.
- **Postprocessing:** Confidence intervals and error bounds are calculated to inform QC staff of prediction reliability.

3.2.4 Real-Time Scenario Analysis

The system includes a real-time simulation feature where plant engineers can run “what-if” analyses.

- Example: Adjusting Blaine fineness from 300 m²/kg to 350 m²/kg shows the impact on 2-day strength.
- Example: Increasing gypsum content from 2.5% to 3.0% demonstrates its influence on early hydration and setting.
- Example: Simulating curing at 30°C vs. 40°C predicts the effect of hot climate conditions on strength gain.

This feature helps the company test multiple process changes before implementing them on the production line.

3.2.5 Output and Reporting System

After predictions are generated, the system presents the results in a clear and actionable format.

- **Outputs:** Predicted compressive strength at 2, 7, and 28 days (MPa), along with confidence levels.

- **Reports:** Automatically generated PDF/Excel reports summarizing inputs, predictions, and recommendations.
- **Interpretation:** The system highlights the most influential factors behind the prediction (e.g., “Low SO₃ may reduce early strength” or “High Blaine improves 2-day strength”).

3.2.6 Notifications and Alerts

The system includes an alert mechanism to notify staff if predictions fall outside acceptable ranges.

- **Examples:**
 - “Warning: Predicted 28-day strength is below standard (≤ 42.5 MPa).”
 - “2-day strength predicted low – consider gypsum or fineness adjustment.”
- **Delivery:** Alerts are sent via the dashboard and optionally as email notifications to QC engineers.

This ensures proactive intervention before a cement batch fails compliance standards.

3 Project Requirements

4.1 Functional Requirements

1. The system should allow laboratory and quality control staff to input cement batch data, including chemical composition, fineness, gypsum/SCM content, environmental factors, and testing parameters.
2. The system should preprocess and clean input data, standardize units, and engineer relevant features for analysis.
3. The system should predict compressive strength at 2, 7, and 28 days for each cement batch using physics-informed and machine learning models.
4. The system should identify and display key influencing factors affecting strength predictions for each batch.
5. The system should enable users to perform real-time scenario analysis by adjusting input parameters and viewing updated strength predictions instantly.
6. The system should provide actionable feedback and recommendations for process adjustments when predicted strength falls below target specifications.
7. The system should generate and display confidence intervals and error margins for all predictions.
8. The system should send notifications and alerts to relevant staff if predicted strength is outside acceptable ranges or if process risks are detected.
9. The system should allow users to download prediction results and reports in Excel or PDF format for record-keeping and analysis.
10. The system should continuously learn from new batch data and laboratory results to improve prediction accuracy over time.

4.2 Non-Functional Requirements

- **Performance:** The system should process and analyze batch data, delivering strength predictions and feedback within 5 seconds.
- **Scalability:** The system should handle increasing numbers of users and growing volumes of production data without significant performance degradation.
- **Reliability:** The system should ensure 99.9% uptime, guaranteeing uninterrupted access to predictions and recommendations, especially during peak production periods.
- **Usability:** The system should provide a user-friendly interface with clear navigation for users of varying technical expertise, including plant and laboratory staff.
- **Security:** The system should protect all user and production data with industry-standard encryption and secure authentication to prevent unauthorized access.
- **Maintainability:** The system should be modular and easy to update, allowing for future enhancements and integration with new data sources or models.

4.3 System Requirements

- Python (for backend and machine learning models)
- Flask or Django (for web application framework)
- React.js or Angular (for frontend dashboard)
- Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow (for data processing and ML)
- PostgreSQL or MySQL (for database management)
- Docker (for containerization and deployment)

- REST API (for integration with plant/lab systems)
- Email/SMS notification service (for alerts)
- Secure server hosting (cloud or on-premises)

4.4. Work Breakdown Structure

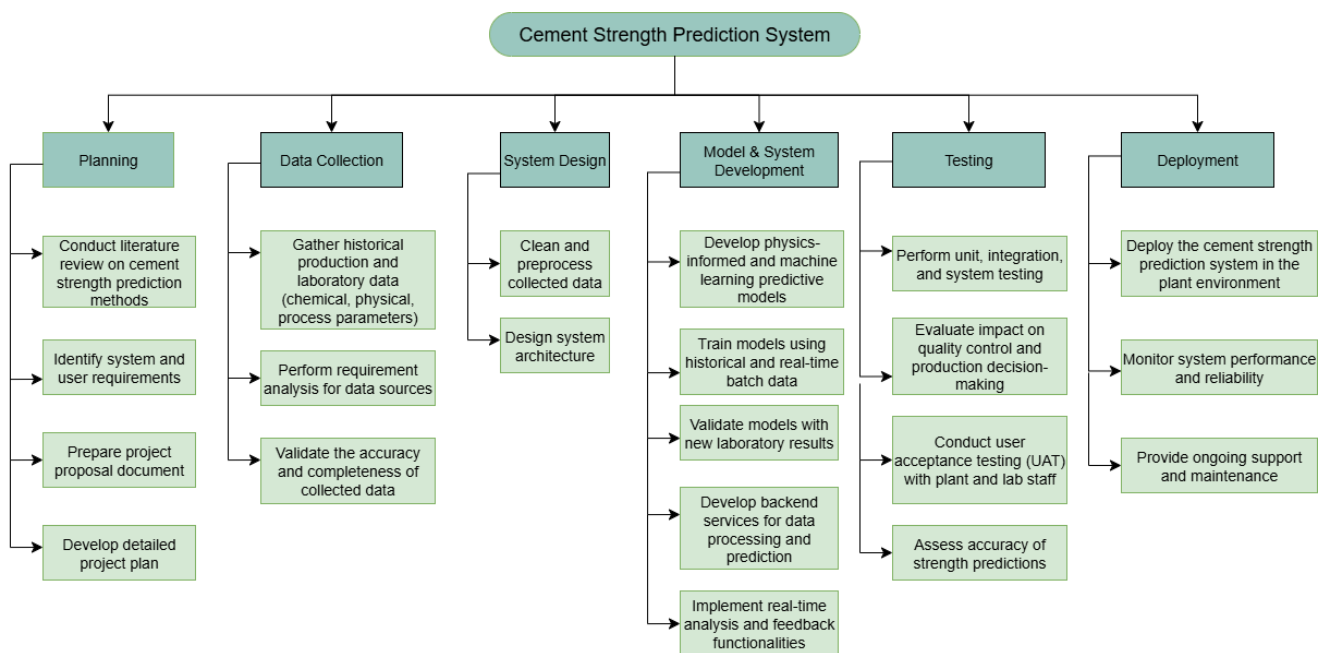


Figure 3: Work Breakdown Structure

5. Budget and Budget justification

The following table represents the estimated budget for the proposed cement strength prediction system.

Requirement	Cost (Rs.)
System Deployment	12,000/month
Weather API	4,500/3-month
Machine Learning Model Maintenance	4,500/3-month
Total	15,000 / month

Table 2: Budget

Budget Justification:

- **System Deployment:** Covers hosting, maintenance, and updates for the web dashboard and backend services.
- **Weather API:** Essential for integrating real-time and historical environmental data. Such as temperature, humidity.
- **Model Maintenance:** Supports ongoing model retraining and accuracy improvements.

6. Commercialization Strategy

The commercialization of the Cement Strength Prediction and Process Optimization System for Sri Lankan cement manufacturers involves strategic steps to ensure adoption, scalability, and sustained revenue.

Main Target Group

- **Cement Manufacturers (e.g., INSEE Cement):**
Primary users who benefit from real-time strength predictions, enabling proactive quality control, process optimization, and reduced production risks.
- **Quality Control and Laboratory Staff:**
Users who input batch data, receive instant feedback, and optimize production parameters.
- **R&D Teams:**
Use the system to evaluate new raw materials and formulations, accelerating product development.

Community Version

- **Essential Tools for Small and Medium Cement Plants:**
 - Batch Data Entry and Strength Prediction: Input batch data and receive compressive strength estimates.
 - Key Factor Analysis: Identify main drivers affecting strength.
 - Basic Reporting: Downloadable reports for record-keeping.
 - Alerts: Receive notifications for batches at risk of failing quality standards.

Premium Version

- **Advanced Features for Large-Scale Operations:**
 - Integration with Plant Automation Systems: Seamless data flow from sensors and lab equipment.

- Scenario Analysis: Test multiple input combinations and process adjustments.
- Customizable Notification Channels: SMS, email, and dashboard alerts.
- Advanced Analytics: Trend analysis, historical performance, and predictive maintenance.
- API Access: For integration with ERP and laboratory information systems.
- Dedicated Support and Model Customization: Tailored solutions for specific production environments.

Market Entry and Scaling

- **Pilot Implementation:**

Deploy the system at INSEE Cement for initial validation and feedback.

- **Partnerships:**

Collaborate with other cement manufacturers and industry associations for broader adoption.

- **Pricing:**

Subscription-based model with tiered pricing for community and premium versions.

Ongoing Development:

Regular updates based on user feedback and evolving industry needs.

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8. Appendices

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